ESAT PRACTICE TEST SERIES

ESAT 2024-25 Recall Mock

Mathematics 1 · Chemistry · Biology

- 27 multiple-choice questions per module
- 40 minutes per module
- Non-calculator
- Hard, recall-informed original practice with full worked solutions

SOLUTION BOOK

ThrivingScholars

ANSWER KEY

ESAT 2024-25 Recall Mock - Mathematics 1

27 hard, recall-informed Mathematics 1 questions with complete worked solutions.

Construction verification: the summary key, marked option, card answer and worked-solution answer have been matched for every question. This is original, unofficial ESAT-style practice informed by the published specification and broad public candidate impressions; it is not an official UAT-UK paper or a reproduction of live-test material.

Mathematics 1 answer key

Q1 **C**	Q2 **A**	Q3 **D**	Q4 **E**	Q5 **D**	Q6 **B**	Q7 **B**	Q8 **C**	Q9 **A**
Q10 **E**	Q11 **C**	Q12 **B**	Q13 **E**	Q14 **E**	Q15 **A**	Q16 **D**	Q17 **A**	Q18 **B**
Q19 **D**	Q20 **A**	Q21 **D**	Q22 **C**	Q23 **E**	Q24 **B**	Q25 **D**	Q26 **E**	Q27 **C**

Mathematics 1

27 hard, recall-informed Mathematics 1 questions with complete worked solutions.

M1-01

Ratio and replacement

Answer C

Question 1

A tank contains an alloy in which copper and zinc are in the mass ratio $5:3$. Exactly $8\ \mathrm{kg}$ of the well-mixed alloy is removed and replaced by $8\ \mathrm{kg}$ of pure zinc. The new copper-to-zinc ratio is $3:2$.

What was the original mass of alloy in the tank?

A $80\ \mathrm{kg}$

B $192\ \mathrm{kg}$

C $200\ \mathrm{kg}$

D $\frac{200}{9}\ \mathrm{kg}$

E $320\ \mathrm{kg}$

WORKED SOLUTION

Let the original mass be M kg. Removing 8 kg of the well-mixed alloy leaves

$$\text{copper} = \frac{5}{8}(M-8), \qquad \text{zinc} = \frac{3}{8}(M-8).$$

After 8 kg of zinc is added,

$$\frac{\frac{5}{8}(M-8)}{\frac{3}{8}(M-8)+8} = \frac{3}{2}.$$

Hence $10(M-8) = 9(M-8) + 192$, so $M = 200$.

Therefore the correct option is C.

Trap check: Option B ignores the removed alloy before the zinc is added; option A treats the removal as changing the copper-to-zinc ratio even though a well-mixed sample preserves it; option D reverses the final ratio; option E treats the removed mass as pure copper.

M1-02

HCF and LCM

Answer **A**

Question 2

For positive integers a and b, the highest common factor of a and b is 6, and their lowest common multiple is 180.

How many *ordered* pairs (a, b) satisfy these conditions?

- **A** 8
- **B** 4
- **C** 6
- **D** 3
- **E** 16

WORKED SOLUTION

Write $a = 6x$ and $b = 6y$, where x and y are coprime. Then

$$6xy = 180 \quad \Longrightarrow \quad xy = 30 = 2 \cdot 3 \cdot 5.$$

Each prime factor must be assigned wholly to either x or y. There are $2^3 = 8$ assignments, and order matters.

Therefore the correct option is A.

Trap check: Option B counts unordered pairs, C omits the assignments with one reduced factor equal to 1, D merely counts the distinct primes, and E incorrectly includes negative ordered pairs despite the positivity condition.

M1-03

Rates from a changing journey

Answer **D**

Question 3

A vehicle travels a route of length D. On one day it covers the first third of the route at speed v and the remainder at speed $2v$, taking 48 minutes in total.

On another day it covers the first half at speed $4v/3$, stops for 6 minutes, and then covers the second half at constant speed kv. The total time is again 48 minutes.

What is k?

- **A** 2
- **B** $\frac{9}{4}$
- **C** $\frac{5}{2}$
- **D** $\frac{12}{5}$
- **E** $\frac{8}{3}$

WORKED SOLUTION

The first journey gives

$$\frac{D/3}{v} + \frac{2D/3}{2v} = \frac{2D}{3v} = 48,$$

so $D/v = 72$ minutes. On the second day the first half takes

$$\frac{D/2}{4v/3} = \frac{3}{8}\frac{D}{v} = 27 \text{ minutes.}$$

After the six-minute stop, 15 minutes remain. Therefore

$$\frac{D/2}{kv} = \frac{36}{k} = 15, \qquad k = \frac{12}{5}.$$

Therefore the correct option is D.

Trap check: Do not average segment speeds without checking the associated times; here the arithmetic mean happens to agree because the two segment times are equal.

M1-04

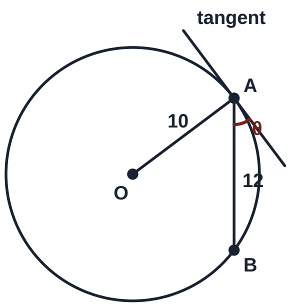
Circle geometry and trigonometry

Answer **E**

Question 4

A circle has centre O, radius 10, and chord AB of length 12. A tangent is drawn at A. Let θ be the acute angle between the tangent and chord AB.

What is $\tan\theta$?



A $\frac{4}{3}$

B $\frac{3}{5}$

C $\frac{4}{5}$

D $\frac{5}{4}$

E $\frac{3}{4}$

WORKED SOLUTION

Let M be the midpoint of AB. The perpendicular from the centre to a chord bisects it, so $AM = 6$. Hence

$$OM = \sqrt{10^2 - 6^2} = 8.$$

If $\alpha = \angle OAB$, then $\tan\alpha = 8/6 = 4/3$. Since the tangent is perpendicular to OA, $\theta = 90^\circ - \alpha$, and therefore

$$\tan\theta = \frac{1}{\tan\alpha} = \frac{3}{4}.$$

Therefore the correct option is E.

Trap check: The value 4/3 is $\tan\angle OAB$, not the tangent–chord angle requested.

M1-05

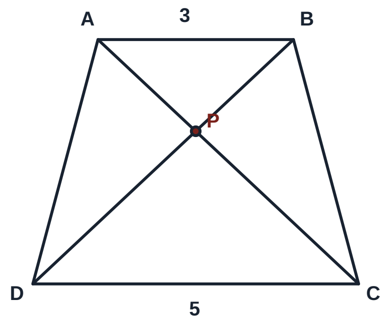
Similarity and area

Answer **D**

Question 5

In trapezium $ABCD$, $AB \parallel CD$. The diagonals AC and BD meet at P, and $AB : CD = 3 : 5$. The area of triangle APB is 27.

What is the area of triangle CPD?



A 45

B 125

C $\frac{243}{25}$

D 75

E 192

WORKED SOLUTION

Triangles APB and CPD are similar: they have equal vertically opposite angles at P, and corresponding angles from $AB \parallel CD$. Their linear scale factor is $3 : 5$, so their areas are in the ratio $9 : 25$.

$$27 : \text{area}(CPD) = 9 : 25.$$

Therefore $\text{area}(CPD) = 27 \times 25/9 = 75$.

Therefore the correct option is D.

Trap check: Option A uses the linear scale factor, B cubes it, C reverses the area factor, and E gives the area of the whole trapezium rather than triangle CPD.

M1-06

Quadratic sequences

Answer **B**

Question 6

The sequence begins

$$7,\ 14,\ 25,\ 40, \ldots$$

and has constant second differences. Let n be the smallest index for which the nth term exceeds 500, and let e be the amount by which that term exceeds 500.

Which pair (n, e) is correct?

- **A** $(15, 31)$
- **B** $(16, 32)$
- **C** $(16, 31)$
- **D** $(15, 32)$
- **E** $(17, 69)$

WORKED SOLUTION

The first differences are $7, 11, 15, \ldots$, so the second difference is 4. Fitting $u_n = 2n^2 + bn + c$ gives

$$u_n = 2n^2 + n + 4.$$

Now $u_{15} = 469$, whereas $u_{16} = 532$. Thus $n = 16$, and the excess is $e = 532 - 500 = 32$.

Therefore the correct option is B.

Trap check: A and D stop one term too early; C subtracts incorrectly after finding the correct index; E moves to the next term after the threshold has already been crossed.

M1-07

Bounds in a quotient

Answer **B**

Question 7

A sample has recorded mass $2.400\ \mathrm{kg}$, correct to the nearest $0.010\ \mathrm{kg}$, and recorded volume $800\ \mathrm{cm}^3$, correct to the nearest $10\ \mathrm{cm}^3$.

If its density is $\rho\ \mathrm{g\ cm}^{-3}$, which statement gives the exact bounds for ρ?

A $\frac{479}{159} \le \rho < \frac{481}{161}$

B $\frac{479}{161} < \rho < \frac{481}{159}$

C $\frac{480}{161} \le \rho < \frac{480}{159}$

D $\frac{2395}{800} \le \rho < \frac{2405}{800}$

E $2.995 \le \rho < 3.005$

WORKED SOLUTION

In grams and cubic centimetres,

$$2395 \le m < 2405, \qquad 795 \le V < 805.$$

For positive quantities, m/V is smallest as m takes its lower bound and V approaches its excluded upper bound; it is greatest as m approaches its excluded upper bound and V takes its lower bound. Hence

$$\frac{2395}{805} < \rho < \frac{2405}{795}.$$

Dividing numerator and denominator by 5 gives

$$\frac{479}{161} < \rho < \frac{481}{159}.$$

Therefore the correct option is B.

Trap check: The lower endpoint is also excluded: the recorded-volume interval has $V < 805$, so density can approach $2395/805$ but cannot equal it.

M1-08

Conditional probability

Answer C

Question 8

A bag contains 4 red, 3 blue and 2 green counters. Two counters are drawn at random without replacement.

Given that the counters are different colours, what is the probability that one of them is green?

A $\frac{5}{13}$

B $\frac{6}{13}$

C $\frac{7}{13}$

D $\frac{7}{18}$

E $\frac{14}{27}$

WORKED SOLUTION

Count unordered pairs. The number of different-colour pairs is

$$4 \cdot 3 + 4 \cdot 2 + 3 \cdot 2 = 26.$$

Of these, the pairs involving green number $4 \cdot 2 + 3 \cdot 2 = 14$. Hence the conditional probability is

$$\frac{14}{26} = \frac{7}{13}.$$

Therefore the correct option is C.

Trap check: The denominator must contain only different-colour pairs, not all $\binom{9}{2}$ pairs.

M1-09

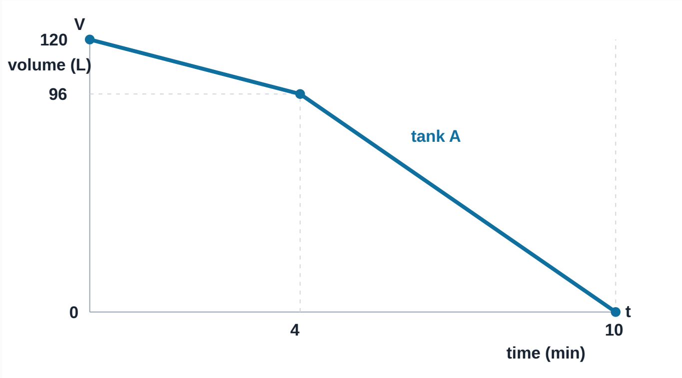
Piecewise linear graphs

Answer **A**

Question 9

The graph shows the volume V litres of water remaining in tank A, t minutes after draining begins. At $t = 0$, an empty tank B starts filling at a constant rate of 12 litres per minute.

At what time do the two tanks contain the same volume?



A $\frac{40}{7}$ min

B 5 min

C 6 min

D $\frac{20}{3}$ min

E 8 min

WORKED SOLUTION

At $t = 4$, tank A contains 96 L. From $t = 4$ to $t = 10$, its volume falls by 96 L in 6 min, so the rate is 16 L/min. On this segment,

$$V_A = 96 - 16(t - 4) = 160 - 16t.$$

Tank B contains $V_B = 12t$. Equating volumes gives

$$160 - 16t = 12t \implies t = \frac{40}{7}.$$

This lies between 4 and 10, so the correct segment was used.

Therefore the correct option is A.

Trap check: The first segment's rule gives a time outside its interval; the change of gradient at 4 minutes matters.

M1-10

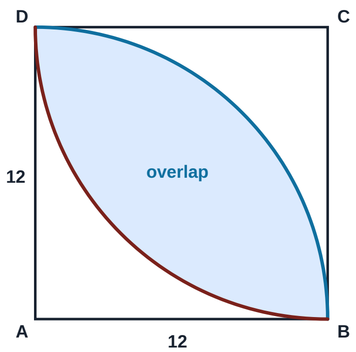
Exact area and sectors

Answer **E**

Question 10

A square $ABCD$ has side length 12. Inside it, quarter-circles of radius 12 are drawn with centres A and C. The shaded lens is the region lying inside both quarter-circles.

What is the exact area of the shaded region?



A $36\pi - 72$

B 72π

C $144 - 36\pi$

D $72\pi - 72$

E $72\pi - 144$

WORKED SOLUTION

Each arc BD subtends 90° at its centre because adjacent sides of the square are perpendicular. One half of the lens is a 90° sector of radius 12, less a right triangle with legs 12:

$$\frac{1}{4}\pi(12)^2 - \frac{1}{2}(12)(12) = 36\pi - 72.$$

The lens has two congruent halves, so its area is

$$2(36\pi - 72) = 72\pi - 144.$$

Therefore the correct option is E.

Trap check: A uses only half the lens, B subtracts no triangles, C takes an incorrect square complement, and D subtracts only one of the two right triangles.

M1-11

Conjugate surds and powers

Answer **C**

Question 11

Let $u = \sqrt{3} + \sqrt{2}$ and $v = \sqrt{3} - \sqrt{2}$.

What is the value of

$$\frac{u^6 + v^6}{u^2 + v^2}\ ?$$

- **A** 10
- **B** 94
- **C** 97
- **D** 100
- **E** 103
- **F** 970

WORKED SOLUTION

Since $uv = 1$ and $u + v = 2\sqrt{3}$,

$$u^2 + v^2 = (u + v)^2 - 2uv = 10.$$

Using $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$ with $a = u^2$, $b = v^2$,

$$u^6 + v^6 = 10^3 - 3(uv)^2(10) = 1000 - 30 = 970.$$

The required quotient is $970/10 = 97$.

Therefore the correct option is C.

Trap check: Option A stops at $u^2 + v^2$, B incorrectly uses $(uv)^2 = 2$, D omits the cubic cross-term, and E changes its sign.

M1-12

Coordinate geometry

Answer **B**

Question 12

The point P lies on the line $y = 2x + 1$ and is equidistant from $A = (0, 4)$ and $B = (6, 0)$. The origin is O.

What is OP^2?

- **A** 169
- **B** 218
- **C** 196
- **D** 224
- **E** 49

WORKED SOLUTION

Write $P = (x, 2x + 1)$. Equating squared distances avoids square roots:

$$x^2 + (2x - 3)^2 = (x - 6)^2 + (2x + 1)^2.$$

The quadratic terms cancel, leaving $-12x + 9 = -8x + 37$. Hence $-4x = 28$, so $x = -7$. Therefore $P = (-7, -13)$, and

$$OP^2 = (-7)^2 + (-13)^2 = 49 + 169 = 218.$$

Therefore the correct option is B.

Trap check: A uses only the square of the y-coordinate, C omits x^2, D introduces a cross-term as if calculating $(x + y)^2$, and E reports only x^2.

M1-13

Restricted sample spaces

Answer **E**

Question 13

Two distinct cards are selected at random from cards numbered 1 to 8. It is known that their product is even and their sum is **not** divisible by 3.

What is the probability that their product is divisible by 4?

A $\frac{9}{22}$

B $\frac{1}{2}$

C $\frac{5}{14}$

D $\frac{7}{11}$

E $\frac{9}{14}$

WORKED SOLUTION

Use unordered pairs. There are 22 pairs with an even product. Of the 10 pairs whose sum is divisible by 3, two have odd product, so eight are excluded from those 22. The conditioned sample space therefore has $22 - 8 = 14$ pairs.

The nine admissible pairs with product divisible by 4 are

$$(2,6),(2,8),(1,4),(3,4),(4,6),(4,7),(6,8),(3,8),(5,8).$$

The required probability is $9/14$.

Therefore the correct option is E.

Trap check: A uses the 22 even-product pairs as the denominator and ignores the sum condition; B assumes an unsupported half-and-half split; C counts only a restricted subset centred on the cards 4 and 6; D counts products divisible by 4 before enforcing the sum condition.

M1-14

Divisors with overlapping conditions

Answer E

Question 14

Let

$$N = 2^4 \cdot 3^3 \cdot 5^2.$$

How many positive divisors of N are divisible by exactly one of 12 and 18?

- A 9
- B 6
- C 18
- D 33
- E 15
- F 51

WORKED SOLUTION

Divisors divisible by $12 = 2^2 \cdot 3$ number

$$(3)(3)(3) = 27.$$

Divisors divisible by $18 = 2 \cdot 3^2$ number

$$(4)(2)(3) = 24.$$

Those divisible by both are divisible by $\operatorname{lcm}(12, 18) = 36 = 2^2 3^2$, giving

$$(3)(2)(3) = 18.$$

To count divisors in exactly one set, subtract the intersection twice:

$$27 + 24 - 2(18) = 15.$$

Therefore the correct option is E.

Trap check: Options A and B count divisors in only one named set separately, C counts the intersection, and D is the ordinary inclusion–exclusion total for at least one condition.

M1-15

Multi-stage ratio

Answer **A**

Question 15

Containers A, B, C initially hold liquid in the ratio $1:3:5$. Half of the liquid in A is transferred to B. Then half of the resulting amount in B is transferred to C. Finally, C contains 100 litres more than A.

How much liquid was there originally in total?

- **A** 144 L
- **B** 200 L
- **C** 150 L
- **D** 180 L
- **E** 108 L

WORKED SOLUTION

Let the initial amounts be $k, 3k, 5k$. After the first transfer,

$$A = \frac{1}{2}k, \qquad B = \frac{7}{2}k.$$

Half of this new amount in B is $7k/4$, so

$$B = \frac{7}{4}k, \qquad C = 5k + \frac{7}{4}k = \frac{27}{4}k.$$

Thus $C - A = (27/4 - 1/2)k = 25k/4 = 100$, giving $k = 16$. The original total was $9k = 144$ L.

Therefore the correct option is A.

Trap check: B ignores the second transfer, C transfers half of B's initial amount, D compares the final amounts in C and B, and E reports C's final amount rather than the original total.

M1-16

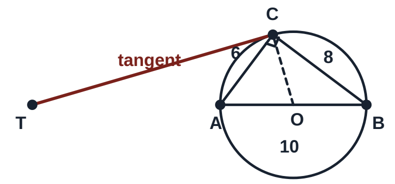
Tangent and coordinate geometry

Answer **D**

Question 16

Triangle ABC is right-angled at C, with $AC = 6$, $BC = 8$ and $AB = 10$. The circle with diameter AB is shown. The tangent to the circle at C meets the line AB, extended beyond A, at T.

What is CT?



A $\frac{60}{7}$

B 12

C 15

D $\frac{120}{7}$

E 20

WORKED SOLUTION

Place $A = (0, 0)$ and $B = (10, 0)$. The 6-8-10 triangle gives $C = (18/5, 24/5)$, and the centre is $O = (5, 0)$. The gradient of OC is

$$\frac{24/5}{18/5 - 5} = -\frac{24}{7}.$$

The tangent is perpendicular to OC, so its gradient is $7/24$. Its equation through C is

$$y - \frac{24}{5} = \frac{7}{24}\left(x - \frac{18}{5}\right).$$

Setting $y = 0$ gives $T = (-90/7, 0)$. Therefore

$$CT = \sqrt{\left(\frac{18}{5} + \frac{90}{7}\right)^2 + \left(\frac{24}{5}\right)^2} = \frac{120}{7}.$$

Therefore the correct option is D.

Trap check: The tangent is perpendicular to the radius OC, not to either side of triangle ABC.

M1-17

Similarity and an inscribed square

Answer **A**

Question 17

A right-angled triangle has perpendicular sides 9 and 12. A square is placed with one side on the hypotenuse and the opposite side parallel to it, with the other two vertices lying on the perpendicular sides.

What is the side length of the square?

- **A** $\frac{180}{37}$
- **B** $\frac{36}{7}$
- **C** $\frac{60}{11}$
- **D** $\frac{72}{13}$
- **E** 6

WORKED SOLUTION

The hypotenuse has length 15, and the altitude from the right angle to the hypotenuse is

$$h = \frac{2 \times (\frac{1}{2})(9)(12)}{15} = \frac{36}{5}.$$

If the square side is s, the small triangle above it is similar to the original triangle. Its scale factor is $(h - s)/h$, so its base, which is a side of the square, satisfies

$$s = 15\frac{h - s}{h}.$$

Thus $s(h + 15) = 15h$, and with $h = 36/5$,

$$s = \frac{15(36/5)}{15 + 36/5} = \frac{180}{37}.$$

Therefore the correct option is A.

Trap check: This is not the standard square placed at the right-angle vertex; here a side of the square lies on the hypotenuse.

M1-18

Statistics and probability

Answer **B**

Question 18

A data set contains only the values shown.

Value	1	2	3	4	5
Frequency	2	a	4	b	2

There are 20 observations and their mean is 3.1. Two observations are selected at random without replacement.

What is the probability that they have equal values?

A $\frac{39}{200}$

B $\frac{39}{190}$

C $\frac{31}{190}$

D $\frac{39}{380}$

E $\frac{3}{10}$

WORKED SOLUTION

The total frequency gives $a+b=12$. The total of all observations is $20(3.1)=62$, so

$$2+2a+12+4b+10=62 \implies a+2b=19.$$

Hence $b=7$ and $a=5$. The number of unordered equal-value pairs is

$$\binom{2}{2}+\binom{5}{2}+\binom{4}{2}+\binom{7}{2}+\binom{2}{2}=1+10+6+21+1=39.$$

There are $\binom{20}{2}=190$ unordered pairs in total, so the probability is $39/190$.

Therefore the correct option is B.

Trap check: Using 20^2 as the denominator models selection with replacement and also permits selecting the same observation twice.

M1-19

Successive variable percentages

Answer **D**

Question 19

A price is increased by $p\%$, where $p > 0$. The increased price is then reduced by $(p+10)\%$. The final price is 12% below the original price.

What is p?

- **A** -20
- **B** 12
- **C** 22
- **D** 10
- **E** 20

WORKED SOLUTION

Let $x = p/100$. The two multipliers give

$$(1+x)(0.9-x) = 0.88.$$

Expanding and simplifying,

$$0.9 - 0.1x - x^2 = 0.88 \quad \Longrightarrow \quad x^2 + 0.1x - 0.02 = 0.$$

Hence $(x-0.1)(x+0.2) = 0$. Since $p > 0$, $x = 0.1$, so $p = 10$.

Therefore the correct option is D.

Trap check: A is the rejected algebraic root, B copies the final percentage, C adds 12 and 10, and E neglects the product term in the two successive multipliers.

M1-20

Prime exponents

Answer **A**

Question 20

What is the smallest positive integer n for which both $72n$ is a perfect square and $50n$ is a perfect cube?

- **A** 20000
- **B** 5000
- **C** 10000
- **D** 16000
- **E** 40000

WORKED SOLUTION

Write $n = 2^a 3^b 5^c$ with the least possible non-negative exponents. Since $72 = 2^3 3^2$, the exponents in $72n$ must be even. Since $50 = 2 \cdot 5^2$, the exponents in $50n$ must be multiples of 3. The least simultaneous choices are

$$a = 5, \qquad b = 0, \qquad c = 4.$$

Thus $n = 2^5 \cdot 5^4 = 32 \cdot 625 = 20000$.

Therefore the correct option is A.

Trap check: The same n must satisfy both exponent conditions simultaneously; satisfying only the square condition is insufficient.

M1-21

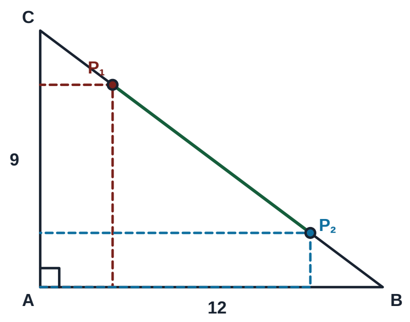
Coordinate geometry and quadratics

Answer **D**

Question 21

Triangle ABC is right-angled at A, with $AB = 12$ and $AC = 9$. For a point P on the hypotenuse, perpendiculars from P to AB and AC form a rectangle having A and P as opposite vertices.

There are two positions, P_1 and P_2, for which the rectangle has area 18. What is P_1P_2?



A $3\sqrt{3}$

B $4\sqrt{3}$

C 5

D $5\sqrt{3}$

E $7\sqrt{3}$

WORKED SOLUTION

Place $A = (0,0)$, $B = (12,0)$ and $C = (0,9)$. If $P = (x,y)$, the hypotenuse equation is $y = 9 - 3x/4$. The rectangle area condition gives

$$x\left(9 - \frac{3}{4}x\right) = 18 \implies x^2 - 12x + 24 = 0.$$

The two roots are $6 \pm 2\sqrt{3}$, so the horizontal separation of P_1, P_2 is $4\sqrt{3}$. Their vertical separation is $(3/4)(4\sqrt{3}) = 3\sqrt{3}$. Hence

$$P_1P_2 = \sqrt{(4\sqrt{3})^2 + (3\sqrt{3})^2} = 5\sqrt{3}.$$

Therefore the correct option is D.

Trap check: The horizontal root difference is only 4√3; P_1P_2 also includes the vertical separation.

M1-22

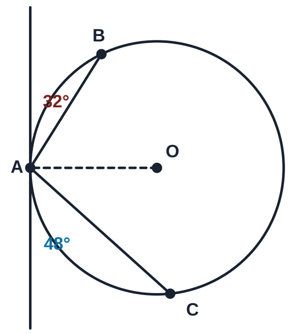
Circle theorems and tangents

Answer **C**

Question 22

At a point A on a circle, a tangent is drawn. Chords AB and AC lie on opposite sides of the radius through A. The acute angles between the tangent and the chords AB and AC are $32°$ and $48°$, respectively.

What is the smaller angle BOC, where O is the centre?



A $100°$

B $80°$

C $160°$

D $200°$

E $40°$

WORKED SOLUTION

By the tangent–chord theorem,

$$\angle ACB = 32°, \qquad \angle ABC = 48°.$$

Therefore $\angle BAC = 180° - 32° - 48° = 100°$. The arc BC not containing A has central angle $200°$. The smaller angle at the centre is therefore

$$360° - 200° = 160°.$$

Therefore the correct option is C.

Trap check: A gives the inscribed angle, B adds the two tangent–chord angles without doubling, D selects the reflex central angle, and E doubles their difference.

M1-23

Sequences and algebraic structure

Answer **E**

Question 23

An increasing arithmetic sequence has positive integer terms. Its terms satisfy

$$u_6^2 = u_3 u_{11} \qquad \text{and} \qquad u_{11} - u_3 = 48.$$

What is the first term of the sequence?

A 6

B 9

C 12

D 18

E 15

WORKED SOLUTION

Let the first term be a and common difference be $d > 0$. Then

$$u_3 = a + 2d, \qquad u_6 = a + 5d, \qquad u_{11} = a + 10d.$$

The difference condition gives $8d = 48$, so $d = 6$. Expanding the other condition,

$$(a + 5d)^2 = (a + 2d)(a + 10d)$$

$$a^2 + 10ad + 25d^2 = a^2 + 12ad + 20d^2,$$

so $5d^2 = 2ad$. Since $d > 0$, $a = 5d/2 = 15$.

Therefore the correct option is E.

Trap check: The identity is multiplicative, not the ordinary arithmetic-mean relation for u_3, u_6 and u_{11}.

M1-24

Conditional probability by structure

Answer **B**

Question 24

Two fair six-sided dice are thrown.

Given that the product of the two scores is divisible by 6, what is the probability that their sum is prime?

A $\frac{7}{15}$

B $\frac{8}{15}$

C $\frac{1}{2}$

D $\frac{3}{5}$

E $\frac{2}{9}$

WORKED SOLUTION

The 15 ordered outcomes whose product is divisible by 6 are

$$(1,6);\ (2,3),(2,6);\ (3,2),(3,4),(3,6);\ (4,3),(4,6);\ (5,6);\ (6,1),(6,2),(6,3),(6,4),(6,5),(6,6).$$

Of these, the eight outcomes

$$(1,6),(6,1),(2,3),(3,2),(3,4),(4,3),(5,6),(6,5)$$

have prime sum. The required probability is therefore $8/15$.

Therefore the correct option is B.

Trap check: A misses one favourable ordered outcome, C counts unordered pairs, D includes one non-prime sum, and E uses the eight favourable outcomes over the original 36-outcome sample space.

M1-25

Congruences with an exclusion

Answer **D**

Question 25

The arithmetic sequence

$$5, 12, 19, 26, \ldots$$

contains terms that are divisible by 11 but not divisible by 3. What is the position of the 15th such term?

- **A** 214
- **B** 225
- **C** 236
- **D** 247
- **E** 258
- **F** 159

WORKED SOLUTION

The nth term is $7n - 2$. Divisibility by 11 requires

$$7n \equiv 2 \pmod{11},$$

so $n \equiv 5 \pmod{11}$. Write $n = 5 + 11k$, where $k \geq 0$. The term is divisible by 3 precisely when $n \equiv 2 \pmod{3}$. Since

$$5 + 11k \equiv 2 + 2k \pmod{3},$$

this happens when $2k \equiv 0 \pmod{3}$, so $k \equiv 0 \pmod{3}$. These values must be excluded. The allowed values of k are therefore

$$1, 2, 4, 5, 7, 8, \ldots$$

There are two allowed values in each block of three. The 15th allowed value is $k = 22$, giving

$$n = 5 + 11(22) = 247.$$

Indeed, $7(247) - 2 = 1727 = 11 \times 157$, which is not divisible by 3.

Therefore the correct option is D.

Trap check: C excludes the wrong residue class of k; F counts the 15th multiple-of-11 candidate without any exclusion; A, B and E are one-block or one-candidate counting shifts.

M1-26

Recurrence and remainders

Answer **E**

Question 26

A positive real number x satisfies

$$x + \frac{1}{x} = 3.$$

For positive integers n, define

$$A_n = x^n + \frac{1}{x^n}.$$

What is the remainder when A_{2025} is divided by 10?

A 2

B 3

C 5

D 7

E 8

WORKED SOLUTION

Multiplying A_n by $x + x^{-1} = 3$ gives

$$3A_n = A_{n+1} + A_{n-1},$$

so $A_{n+1} = 3A_n - A_{n-1}$. Also $A_0 = 2$ and $A_1 = 3$. Working modulo 10,

$$A_0, A_1, A_2, A_3, A_4, A_5, A_6 \equiv 2, 3, 7, 8, 7, 3, 2.$$

The pair $(A_6, A_7) \equiv (2, 3)$ repeats the initial pair, so the remainders have period 6. Since 2025 leaves remainder 3 on division by 6,

$$A_{2025} \equiv A_3 \equiv 8 \pmod{10}.$$

Therefore the correct option is E.

Trap check: Track a pair of consecutive remainders to justify the cycle; matching one term alone is not enough.

M1-27

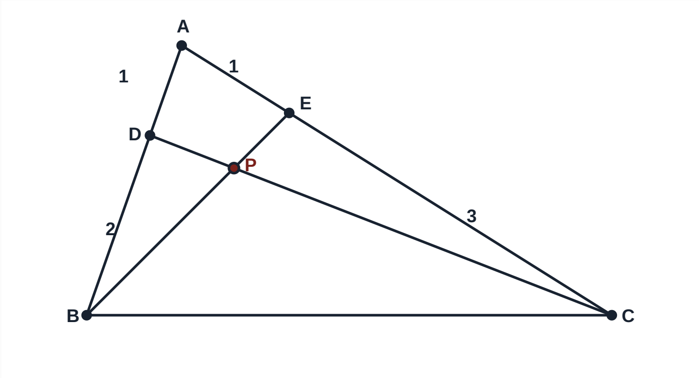
Cevians and coordinate reasoning

Answer **C**

Question 27

In triangle ABC, point D lies on AB with $AD : DB = 1 : 2$, and point E lies on AC with $AE : EC = 1 : 3$. Lines DC and EB intersect at P.

What is $EP : PB$?



A $3 : 7$

B $8 : 3$

C $3 : 8$

D $4 : 9$

E $2 : 7$

F $3 : 11$

G $11 : 3$

H $8 : 11$

WORKED SOLUTION

Choose convenient coordinates $A = (0, 0)$, $B = (3, 0)$, $C = (0, 4)$. Then $D = (1, 0)$ and $E = (0, 1)$. Parameterise P on both cevians:

$$P = D + t(C - D) = (1 - t, 4t),$$

$$P = E + s(B - E) = (3s, 1 - s).$$

Solving $1 - t = 3s$ and $4t = 1 - s$ gives $s = 3/11$. Since s is the fraction of the way from E to B,

$$EP : PB = s : (1 - s) = \frac{3}{11} : \frac{8}{11} = 3 : 8.$$

Therefore the correct option is C.

Trap check: B reverses the required ratio; F reports $EP : EB = 3 : 11$; G reverses that whole-segment ratio; H reports $PB : EB = 8 : 11$. A, D and E arise from using one of the side ratios directly or mismatching coefficients in the two line parameters.

ANSWER KEY

ESAT 2024-25 Recall Mock - Chemistry

27 hard, recall-informed Chemistry questions with complete worked solutions.

Construction verification: the summary key, marked option, card answer and worked-solution answer have been matched for every question. This is original, unofficial ESAT-style practice informed by the published specification and broad public candidate impressions; it is not an official UAT-UK paper or a reproduction of live-test material.

Chemistry answer key

Q1 C	Q2 A	Q3 E	Q4 B	Q5 D	Q6 B	Q7 A	Q8 D	Q9 C
Q10 E	Q11 C	Q12 A	Q13 E	Q14 B	Q15 D	Q16 A	Q17 C	Q18 E
Q19 B	Q20 D	Q21 E	Q22 C	Q23 A	Q24 D	Q25 B	Q26 C	Q27 A

Chemistry

27 hard, recall-informed Chemistry questions with complete worked solutions.

C-01

Isotopes and probability

Answer **C**

Question 1

Element X has two naturally occurring isotopes, ^{63}X and ^{65}X. Its relative atomic mass is 63.5. Atoms combine at random to form X_2 molecules. What percentage of X_2 molecules are expected to have relative molecular mass 128?

- **A** 12.5%
- **B** 25.0%
- **C** 37.5%
- **D** 50.0%
- **E** 75.0%

WORKED SOLUTION

Let the fraction of ^{65}X be p. Then $63(1-p)+65p=63.5$, so $p=0.25$ and the ^{63}X fraction is 0.75. Mass 128 requires one atom of each isotope, in either order: $2(0.75)(0.25)=0.375$ = 37.5%.

Therefore the correct option is **C**.

C-02

Ions, formulae and limiting amounts

Answer **A**

Question 2

Atoms P and Q have electron configurations P: 2,8,2 and Q: 2,7. In a two-stage process, 0.240 mol of P atoms and 0.360 mol of Q atoms first react as completely as possible. The ionic solid is removed, and any unreacted P is then added to excess acid:

$P + 2H^+ \rightarrow P^{2+} + H_2$

Which option correctly gives the ionic solid formed in the first stage and the amount of hydrogen formed in the second?

- **A** 0.180 mol of PQ_2; 0.060 mol H_2
- **B** 0.240 mol of PQ_2; no H_2
- **C** 0.120 mol of P_2Q; 0.120 mol H_2
- **D** 0.180 mol of covalent PQ_2; 0.060 mol H_2
- **E** 0.360 mol of PQ_2; 0.120 mol H_2

WORKED SOLUTION

P forms P^{2+} and Q forms Q^-, so the ionic formula is PQ_2. The 0.360 mol of Q atoms can form 0.360/2=0.180 mol PQ_2, using 0.180 mol P. This leaves 0.240−0.180=0.060 mol P. The acid equation has a 1:1 ratio between P and H_2, so the second stage forms 0.060 mol H_2.

Therefore the correct option is **A**.

Distractor check: B ignores the 1:2 formula ratio and the leftover P; C reverses the ionic formula and doubles the gas; D has the right amounts but the wrong bonding type; E equates the amount of Q atoms with the amount of compound.

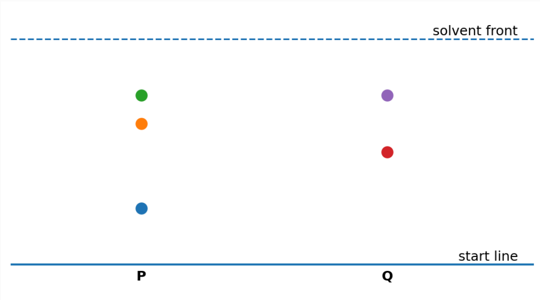
C-03

Chromatography and Rf reasoning

Answer **E**

Question 3

The chromatograms P and Q were run with the same solvent and stationary phase. The solvent front travelled 8.0 cm in each case. In P, spot distances from the start are 2.0, 5.0 and 6.0 cm. In Q, they are 4.0 and 6.0 cm. Reference substances have R_f values W=0.25, X=0.50, Y=0.625 and Z=0.75. Which statement is correct?



A P contains W, X and Y

B P and Q both contain X

C Only Q contains Z

D Q contains W and Y

E P contains W, Y and Z, while Q contains X and Z

WORKED SOLUTION

For P, R_f values are 0.25, 0.625 and 0.75, so P contains W, Y and Z. For Q they are 0.50 and 0.75, so Q contains X and Z.

Therefore the correct option is **E**.

C-04

Limiting reagent and gas volume

Answer **B**

Question 4

2.40 g of magnesium is added to 100 cm^3 of 1.00 mol dm^{-3} hydrochloric acid. When gas production stops, the unreacted magnesium is separated without loss and added to 100 cm^3 of 0.500 mol dm^{-3} hydrochloric acid.

$Mg + 2HCl \rightarrow MgCl_2 + H_2$

Take $A_r(Mg)=24$ and molar gas volume at rtp as 24 dm^3 mol^{-1}. What total volume of hydrogen is produced across the two stages?

- **A** 0.60 dm^3
- **B** 1.80 dm^3
- **C** 2.40 dm^3
- **D** 1.20 dm^3
- **E** 3.60 dm^3

WORKED SOLUTION

Initially there are 2.40/24=0.100 mol Mg and 0.100 mol HCl. The first acid portion therefore forms 0.050 mol H_2 and consumes 0.050 mol Mg, leaving 0.050 mol Mg. The second acid portion contains 0.100×0.500=0.050 mol HCl, so it forms a further 0.025 mol H_2. The total is 0.075 mol, occupying 0.075×24=1.80 dm^3 at rtp.

Therefore the correct option is **B**.

Distractor check: A counts only the second stage; C assumes all the original magnesium reacts; D counts only the first stage; E treats the HCl:H_2 ratio as 1:1 in both stages.

C-05

Equilibrium and rate

Answer **D**

Question 5

For $N_2(g)+3H_2(g) \rightleftharpoons 2NH_3(g)$, the forward reaction is exothermic. Which single change increases both the equilibrium yield of ammonia and the initial forward reaction rate?

- **A** decrease temperature
- **B** increase temperature
- **C** add a catalyst
- **D** decrease the container volume
- **E** remove the catalyst

WORKED SOLUTION

Decreasing volume increases pressure, favouring the side with fewer gas molecules (the products), and also increases collision frequency. A catalyst increases rate but not equilibrium yield; lowering temperature improves yield but slows the reaction.

Therefore the correct option is **D**.

C-06

Disproportionation and oxidation states

Answer **B**

Question 6

In stage 1, 0.030 mol Cl_2 reacts completely with excess hydroxide ions:

$Cl_2 + 2OH^- \rightarrow Cl^- + ClO^- + H_2O$

The entire solution is then acidified and mixed with excess iodide ions. In stage 2, all the ClO^- reacts:

$ClO^- + 2I^- + 2H^+ \rightarrow Cl^- + I_2 + H_2O$

Which option gives the final amounts of Cl^- and I_2, and the role of Cl_2 in stage 1?

A 0.030 mol Cl^-; 0.030 mol I_2; both oxidising and reducing agent

B 0.060 mol Cl^-; 0.030 mol I_2; both oxidising and reducing agent

C 0.060 mol Cl^-; 0.060 mol I_2; oxidising agent only

D 0.030 mol Cl^-; 0.060 mol I_2; reducing agent only

E 0.090 mol Cl^-; 0.030 mol I_2; neither

WORKED SOLUTION

Stage 1 forms 0.030 mol Cl^- and 0.030 mol ClO^-. Stage 2 converts the 0.030 mol ClO^- into a further 0.030 mol Cl^- and forms 0.030 mol I_2. The final chloride amount is therefore 0.060 mol. In stage 1, chlorine changes from oxidation state 0 to −1 in Cl^- and to +1 in ClO^-, so Cl_2 acts as both oxidising and reducing agent.

Therefore the correct option is **B**.

Distractor check: A omits the chloride made in stage 2; C doubles the iodine and ignores disproportionation; D reverses both the second-stage ratio and the redox role; E counts three chloride-producing routes although only two occur.

C-07

Structure, bonding and evidence

Answer A

Question 7

Two pure solids are tested.

solid	melting point	electrical conduction	mechanical behaviour	water solubility
R	high	solid and molten	bends without shattering	insoluble
S	high	molten only	shatters when struck	soluble

Which explanation best fits all the evidence?

- **A** R is metallic, with delocalised electrons; S is ionic, with ions mobile only when molten.
- **B** R is ionic; S is metallic.
- **C** R is graphite; S is ionic.
- **D** R is metallic; S is giant covalent.
- **E** R is simple molecular; S is ionic.

WORKED SOLUTION

R's high melting point, conduction in both states and ability to bend are consistent with a metal: delocalised electrons carry charge and layers of positive ions can slide while metallic bonding persists. S is consistent with an ionic solid: its ions are fixed in the solid but mobile in the melt; brittleness and water solubility provide supporting evidence.

Therefore the correct option is **A**.

Distractor check: B reverses the conductivity and mechanical patterns. C fits several observations but graphite is brittle rather than malleable. D fits R completely and some of S, but a typical giant covalent solid neither dissolves in water nor conducts when molten. E fits S but conflicts with R's high melting point and conduction.

C-08

Halogen displacement

Answer D

Question 8

A halogen X_2 does not displace chloride ions from solution, but does displace bromide and iodide ions. Which statement is correct?

- **A** X is bromine; X_2 is oxidised when it reacts with bromide ions.
- **B** X is chlorine; bromine can oxidise X^- ions.
- **C** X is fluorine; X_2 is reduced only when it reacts with iodide ions.
- **D** X is chlorine; X_2 is reduced when it reacts with bromide or iodide ions.
- **E** X is bromine; iodine can oxidise X^- ions.

WORKED SOLUTION

The displacement pattern identifies X as chlorine: chlorine does not displace chloride ions, but it is more reactive than bromine and iodine and so oxidises Br^- and I^-. In each displacement reaction, Cl_2 gains electrons to form Cl^-, so X_2 is reduced. Bromine cannot oxidise chloride ions because bromine is less reactive than chlorine.

Therefore the correct option is **D**.

C-09

Rates and controlled comparisons

Answer **C**

Question 9

In each experiment, the same amount of calcium carbonate reacts with the same volume of excess hydrochloric acid.

experiment	acid concentration / mol dm^{-3}	temperature / °C	carbonate form	time to collect 40 cm^3 CO_2 / s	final CO_2 volume / cm^3
X	1.0	20	chips	80	48
Y	2.0	20	chips	45	48
Z	1.0	30	powder	20	48

Which conclusion is justified by the data?

A Y produces a greater final amount of carbon dioxide than X.

B Z proves that raising temperature alone makes the reaction faster.

C X and Y support the conclusion that higher acid concentration increases rate but not the final amount when carbonate is limiting.

D Equal final volumes show that all three reactions have the same rate.

E Z must contain the least calcium carbonate because it reaches 40 cm^3 first.

WORKED SOLUTION

X and Y differ only in acid concentration. Y reaches 40 cm^3 sooner, so it has a higher rate, while both reach 48 cm^3 because the same amount of limiting calcium carbonate reacts. X and Z are not a controlled comparison for temperature because particle size also changes.

Therefore the correct option is **C**.

Distractor check: A contradicts the final volumes; B ignores the simultaneous surface-area change; D confuses amount with rate; E confuses reaching a volume sooner with producing less gas overall.

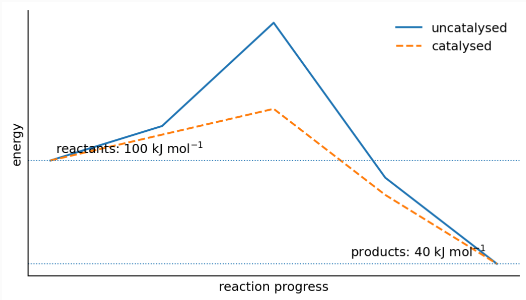
C-10

Energy profile and catalysts

Answer **E**

Question 10

The energy profile shows reactants at 100 kJ mol^{-1}, products at 40 kJ mol^{-1}, and a catalysed peak at 130 kJ mol^{-1}. What are ΔH for the forward reaction and the activation energy of the catalysed reverse reaction?



A −60 and 30 kJ mol^{-1}

B +60 and 30 kJ mol^{-1}

C +60 and 90 kJ mol^{-1}

D −30 and 60 kJ mol^{-1}

E −60 and 90 kJ mol^{-1}

WORKED SOLUTION

$\Delta H = 40-100 = -60$ kJ mol^{-1}. For the catalysed reverse reaction, $E_a=130-40=90$ kJ mol^{-1}.

Therefore the correct option is **E**.

C-11

Bond-energy inference

Answer **C**

Question 11

The following gaseous reactions have the stated enthalpy changes:

$H_2 + Cl_2 \rightarrow 2HCl \quad \Delta H = -184\ kJ\ mol^{-1}$

$H_2 + Br_2 \rightarrow 2HBr \quad \Delta H = -102\ kJ\ mol^{-1}$

The average bond energies are H−H=436, Cl−Cl=242 and Br−Br=194 kJ mol^{-1}. By how much is the average H−Cl bond energy greater than the average H−Br bond energy?

- **A** 32.5 kJ mol^{-1}
- **B** 41 kJ mol^{-1}
- **C** 65 kJ mol^{-1}
- **D** 82 kJ mol^{-1}
- **E** 130 kJ mol^{-1}

WORKED SOLUTION

For HCl, −184=(436+242)−2E(H−Cl), so E(H−Cl)=862/2=431 kJ mol^{-1}. For HBr, −102=(436+194)−2E(H−Br), so E(H−Br)=732/2=366 kJ mol^{-1}. The difference is 431−366=65 kJ mol^{-1}.

Therefore the correct option is **C**.

Distractor check: A halves the final difference again; B uses only half the difference between the reaction enthalpies; D reports the reaction-enthalpy difference; E forgets that two H−X bonds form in each reaction.

C-12

pH and dilution

Answer **A**

Question 12

A strong acid has pH 2.00. A 10.0 cm^3 portion is diluted with water to 100.0 cm^3. Then 20.0 cm^3 of this diluted acid is mixed with 180 cm^3 of 1.00×10^{-4} mol dm^{-3} sodium hydroxide.

Assume complete dissociation and neutralisation, additive volumes and negligible ionisation of water. What is the final pH?

- **A** 5.00
- **B** 3.00
- **C** 4.00
- **D** 6.00
- **E** 7.00

WORKED SOLUTION

The original acid has $[H^+] = 1.00 \times 10^{-2}$ mol dm^{-3}. Tenfold dilution makes this 1.00×10^{-3} mol dm^{-3}. The 20.0 cm^3 portion contains 2.00×10^{-5} mol H^+, while the alkali contains $0.180 \times 1.00 \times 10^{-4} = 1.80 \times 10^{-5}$ mol OH^-. After neutralisation, 2.00×10^{-6} mol H^+ remains in 0.200 dm^3, giving $[H^+] = 1.00 \times 10^{-5}$ mol dm^{-3} and pH 5.00.

Therefore the correct option is **A**.

Distractor check: B stops after the first dilution; C treats the mixing as dilution only and ignores hydroxide; D uses 2.00 dm^3 instead of 0.200 dm^3 as the final volume; E assumes exact neutralisation.

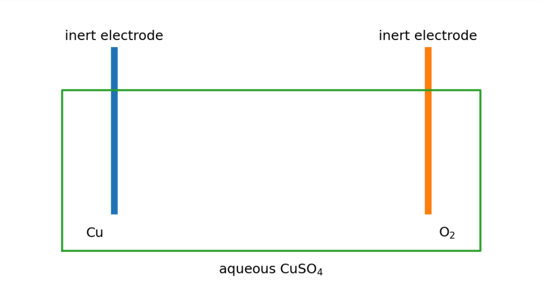
C-13

Electrolysis and stoichiometry

Answer **E**

Question 13

Aqueous copper(II) sulfate is electrolysed with inert electrodes. Assume Cu^{2+} ions remain in excess throughout and that no other electrode reaction occurs. If 0.025 mol of oxygen is formed at the anode, what mass of copper is deposited at the cathode? Take $A_r(Cu)=63.5$.



A 1.59 g

B 6.35 g

C 12.7 g

D 25.4 g

E 3.18 g

WORKED SOLUTION

Forming 1 mol O_2 corresponds to 4 mol electrons, so 0.025 mol O_2 corresponds to 0.100 mol e^-. Cu^{2+} needs 2 e^- per Cu, so 0.050 mol Cu forms: $0.050 \times 63.5 = 3.175$ g ≈ 3.18 g.

Therefore the correct option is **E**.

C-14

Neutralisation and stoichiometry

Answer **B**

Question 14

25.0 cm^3 of 0.200 mol dm^{-3} sulfuric acid is mixed with 40.0 cm^3 of 0.250 mol dm^{-3} sodium hydroxide. Assume complete neutralisation and additive volumes.

$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$

A 13.0 cm^3 aliquot of the resulting 65.0 cm^3 mixture is removed. Then 5.0 cm^3 of 0.200 mol dm^{-3} sodium hydroxide is added to what remains. Which statement describes the final mixture?

- **A** It is neutral and contains 0.0040 mol of Na_2SO_4.
- **B** It is alkaline and contains 0.0040 mol of Na_2SO_4.
- **C** It is alkaline and contains 0.0050 mol of Na_2SO_4.
- **D** It is neutral and contains 0.0050 mol of Na_2SO_4.
- **E** It is acidic and contains 0.0010 mol of Na_2SO_4.

WORKED SOLUTION

Initially there are 0.00500 mol H_2SO_4 and 0.0100 mol NaOH, so they react exactly and form 0.00500 mol Na_2SO_4. Removing 13.0/65.0=1/5 of the homogeneous mixture removes 1/5 of the salt, leaving 0.00400 mol. The added sodium hydroxide is now in excess because no acid remains, so the final mixture is alkaline; it does not change the amount of Na_2SO_4.

Therefore the correct option is **B**.

Distractor check: A ignores the final sodium hydroxide; C ignores removal of the aliquot; D ignores both later operations; E reports the removed salt amount and reverses the acid–base conclusion.

C-15

Hydrocarbon data and addition

Answer **D**

Question 15

A 0.280 g sample of gaseous hydrocarbon X occupies 240 cm^3 at rtp. X decolourises bromine water. Take the molar gas volume at rtp as 24 $dm^3\ mol^{-1}$.

Which option gives the formula of X and the formula of the organic product when X reacts with bromine?

- **A** CH_4; CH_3Br
- **B** C_2H_6; $C_2H_6Br_2$
- **C** C_2H_4; C_2H_5Br
- **D** C_2H_4; $C_2H_4Br_2$
- **E** C_3H_6; $C_3H_6Br_2$

WORKED SOLUTION

The gas amount is 0.240/24 = 0.010 mol, so $M_r(X)=0.280/0.010=28$. The hydrocarbon with M_r 28 is C_2H_4. Its decolourisation of bromine water is consistent with a C=C bond, and bromine adds across that bond to form $C_2H_4Br_2$.

Therefore the correct option is **D**.

Distractor check: C treats addition as substitution; E uses the formula of the next alkene without reconciling it with the measured molar mass.

C-16

Thermal decomposition data

Answer **A**

Question 16

A 10.0 g sample of a metal carbonate is heated, but the decomposition is stopped before completion:

$MCO_3(s) \rightarrow MO(s) + CO_2(g)$

The heating produces 1.92 dm^3 CO_2 at rtp. The cooled solid is then treated with excess acid, producing a further 0.48 dm^3 CO_2 at rtp. In the acid, each mole of undecomposed MCO_3 produces one mole of CO_2, while MO produces no gas. Take molar gas volume=24 $dm^3\ mol^{-1}$, $A_r(C)=12$ and $A_r(O)=16$. What is $A_r(M)$?

- **A** 40
- **B** 65
- **C** 100
- **D** 440
- **E** 160

WORKED SOLUTION

Each mole of MCO_3 produces one mole of CO_2, whether it decomposes during heating or reacts later with acid. The total gas volume is 1.92+0.48=2.40 dm^3, corresponding to 2.40/24=0.100 mol MCO_3. Therefore $M_r(MCO_3)=10.0/0.100=100$, so $A_r(M)=100-(12+3\times16)=40$.

Therefore the correct option is **A**.

Distractor check: B uses only the gas made during heating; C reports the relative formula mass of MCO_3; D uses only the undecomposed amount; E adds the carbonate-group mass instead of subtracting it.

C-17

Structural isomerism in alkenes

Answer **C**

Question 17

Two straight-chain compounds have condensed structural formulae CH_2=$CHCH_2CH_3$ and CH_3CH=$CHCH_3$.

Which description is correct?

- **A** They are the same compound written from opposite ends.
- **B** They have different molecular formulae but belong to the same homologous series.
- **C** They are structural isomers and both have molecular formula C_4H_8.
- **D** They have the same molecular formula but different functional groups.
- **E** They are chain isomers and one is saturated.

WORKED SOLUTION

Both structures contain four carbon atoms, eight hydrogen atoms and one C=C bond, so both have molecular formula C_4H_8. The double bond is in a different position, so the displayed connectivity differs: they are structural (positional) isomers.

Therefore the correct option is **C**.

Distractor check: A confuses reversal of a formula with movement of the double bond; D overlooks that both compounds are alkenes.

C-18

Metals and extraction

Answer **E**

Question 18

Metal M is below carbon in the reactivity series but above hydrogen. Consider the statements.
1. An oxide of M can be reduced by carbon under suitable conditions.
2. M reacts with dilute hydrochloric acid to release hydrogen.
3. During electrolysis of an aqueous salt of M using inert electrodes, M is deposited at the cathode in preference to hydrogen.
Which statements are correct?

- **A** 1 only
- **B** 1 and 3 only
- **C** 2 and 3 only
- **D** 1, 2 and 3
- **E** 1 and 2 only

WORKED SOLUTION

Because M is below carbon, carbon can reduce its oxide under suitable conditions, so statement 1 is correct. Because M is above hydrogen, it can displace hydrogen from dilute acids, so statement 2 is correct. But for an aqueous electrolyte containing ions of a metal above hydrogen, hydrogen is discharged at the cathode in preference to the metal, so statement 3 is false.

Therefore the correct option is **E**.

C-19

Chemical analysis and interference

Answer **B**

Question 19

An aqueous salt gives a green precipitate with sodium hydroxide solution. A separate sample is acidified with dilute hydrochloric acid and then gives a white precipitate with barium chloride solution. Which inference is best?

A The salt contains Fe^{2+} and Cl^-; acid is added to make the green precipitate form.

B The salt contains Fe^{2+} and SO_4^{2-}; acidification prevents carbonate ions giving a misleading barium precipitate.

C The salt contains Cu^{2+} and SO_4^{2-}; acid changes a blue precipitate to green.

D The salt contains Mg^{2+} and SO_4^{2-}; barium chloride identifies the cation.

E The salt contains Fe^{2+} and NO_3^-; hydrochloric acid supplies the sulfate ions.

WORKED SOLUTION

The green hydroxide precipitate indicates Fe^{2+}. A white precipitate after acidification and addition of barium chloride indicates sulfate ions. Acidification removes carbonate ions as carbon dioxide, preventing insoluble barium carbonate from causing a false positive.

Therefore the correct option is **B**.

Distractor check: A assigns the wrong anion and purpose to the acid; C and D mismatch the hydroxide colours; E wrongly claims hydrochloric acid contains sulfate.

C-20

Catalytic converter: redox and gas volumes

Answer **D**

Question 20

A 160 cm^3 exhaust pulse contains 90 cm^3 CO, 60 cm^3 NO and 10 cm^3 O_2. In a model catalytic converter, reaction 1 goes to completion before reaction 2 occurs:

1 $2CO + 2NO \rightarrow 2CO_2 + N_2$
2 $2CO + O_2 \rightarrow 2CO_2$

All gas volumes are measured under the same conditions. Which option gives the final total gas volume and identifies the oxidising agent in each reaction?

A 90 cm^3; reaction 1: CO; reaction 2: CO

B 110 cm^3; reaction 1: NO; reaction 2: CO

C 120 cm^3; reaction 1: CO; reaction 2: NO

D 120 cm^3; reaction 1: NO; reaction 2: O_2

E 150 cm^3; reaction 1: O_2; reaction 2: O_2

WORKED SOLUTION

Reaction 1 consumes 60 cm^3 CO and all 60 cm^3 NO, forming 60 cm^3 CO_2 and 30 cm^3 N_2; 30 cm^3 CO remains. Reaction 2 uses all 10 cm^3 O_2 and 20 cm^3 of that CO, forming another 20 cm^3 CO_2. The final gases are 10 cm^3 CO, 80 cm^3 CO_2 and 30 cm^3 N_2, totalling 120 cm^3. NO is reduced in reaction 1 and O_2 is reduced in reaction 2, so each is the oxidising agent in its reaction.

Therefore the correct option is **D**.

Distractor check: A treats the reducing agent as the oxidising agent in both stages; B identifies the first agent but loses a product volume and misclassifies CO in reaction 2; C finds the volume but reverses the redox roles; E ignores the first-stage reduction of NO and both gas contractions.

C-21

Solubility and crystallisation

Answer **E**

Question 21

A salt has solubility 80 g per 100 g water at 80°C and 20 g per 100 g water at 20°C. A 270 g saturated solution at 80°C is kept at 80°C while 30 g of water evaporates; any crystals formed remain in the vessel. The contents are then cooled to 20°C with no further water loss. What total mass of salt is present as crystals at 20°C?

A 24 g

B 72 g

C 90 g

D 120 g

E 96 g

WORKED SOLUTION

At 80°C, 180 g saturated solution contains 100 g water and 80 g salt, so 270 g contains 150 g water and 120 g salt. Evaporation leaves 120 g water. At 80°C this can keep 96 g salt dissolved, so 24 g has already crystallised. At 20°C, 120 g water can hold only 24 g salt, so the total crystal mass is 120−24=96 g.

Therefore the correct option is **E**.

Distractor check: A gives either the salt crystallised during evaporation or the salt left dissolved at 20°C; B counts only the additional crystallisation on cooling; C ignores the water loss; D assumes all salt crystallises.

C-22

Gas volumes and equations

Answer **C**

Question 22

At the same temperature and pressure, 30 cm^3 of gaseous A reacts exactly with 45 cm^3 of gaseous B to form 30 cm^3 of gaseous C only. Which equation has the simplest whole-number coefficients consistent with the data?

A A + 1.5B → C

B 2A + 3B → 3C

C 2A + 3B → 2C

D 3A + 2B → 3C

E 4A + 6B → 4C

WORKED SOLUTION

At the same temperature and pressure, gas volumes are proportional to amounts in moles. The ratio 30:45:30 simplifies to 2:3:2. Therefore the simplest whole-number equation is 2A + 3B → 2C. Option E has the same ratio but its coefficients are not in simplest form.

Therefore the correct option is **C**.

C-23

Calorimetry with energy transfer

Answer **A**

Question 23

A student burns 0.010 mol of a fuel. The combustion raises the temperature of 200 g of water by 12.5°C. The specific heat capacity of water is 4.2 J g^{-1} °C^{-1}.

Assume that 70% of the energy released by the fuel is transferred to the water. What is the molar energy release of the fuel?

- **A** 1500 kJ mol^{-1}
- **B** 1050 kJ mol^{-1}
- **C** 735 kJ mol^{-1}
- **D** 150 kJ mol^{-1}
- **E** 15 kJ mol^{-1}

WORKED SOLUTION

The water gains $q=mc\Delta T=200\times4.2\times12.5=10\,500$ J. This is 70% of the fuel's energy release, so the fuel releases 10 500/0.70=15 000 J for 0.010 mol. Per mole this is 1 500 000 J mol^{-1}=1500 kJ mol^{-1}.

Therefore the correct option is **A**.

Distractor check: B ignores the 70% transfer; C applies the efficiency in the wrong direction; D and E lose powers of ten during the per-mole or joule-to-kilojoule conversion.

C-24

Sulfur dioxide emissions data

Answer **D**

Question 24

On complete combustion, assume that all sulfur in a fuel is converted to sulfur dioxide.

fuel	mass of sulfur burned	energy released
P	0.80 g	40 MJ
Q	0.30 g	10 MJ

Take $A_r(S)=32$ and $A_r(O)=16$. Which row gives the sulfur dioxide emitted per MJ and identifies the lower-emission fuel?

- **A** P: 0.020 g MJ^{-1}; Q: 0.030 g MJ^{-1}; P is lower
- **B** P: 0.040 g MJ^{-1}; Q: 0.060 g MJ^{-1}; Q is lower
- **C** P: 0.080 g MJ^{-1}; Q: 0.120 g MJ^{-1}; P is lower
- **D** P: 0.040 g MJ^{-1}; Q: 0.060 g MJ^{-1}; P is lower
- **E** P: 0.040 g MJ^{-1}; Q: 0.030 g MJ^{-1}; Q is lower

WORKED SOLUTION

$M_r(SO_2)=64$, so each gram of sulfur forms 64/32=2 g of SO_2. Fuel P therefore emits 1.60/40=0.040 g MJ^{-1}; fuel Q emits 0.60/10=0.060 g MJ^{-1}. P has the lower emission per unit energy.

Therefore the correct option is **D**.

Distractor check: A compares sulfur rather than sulfur dioxide; B calculates both rates but reverses the comparison; C applies the mass factor twice.

C-25

Separation and product recovery

Answer **B**

Question 25

A mixture contains water, dissolved sodium chloride and insoluble sand. Which sequence can produce both pure water and dry sodium chloride as separate products?

A Filter the mixture, then evaporate the filtrate to dryness while allowing the water vapour to escape.

B Filter the mixture. Distil the filtrate and collect the water, stopping before the residue becomes dry. Evaporate the remaining concentrated solution and dry the sodium chloride crystals.

C Distil the original mixture and then dry its residue, without first separating the sand.

D Filter the mixture, crystallise some sodium chloride, and use the remaining mother liquor as the pure-water product.

E Centrifuge the mixture and decant the liquid as pure water, then dry the sediment as sodium chloride.

WORKED SOLUTION

Filtration removes the sand. Simple distillation produces pure water as the distillate. The remaining concentrated sodium chloride solution is then evaporated and the salt is dried.

Therefore the correct option is **B**.

Distractor check: A recovers salt but not water; C leaves sand mixed with the salt; D mistakes salt solution for pure water; E cannot remove dissolved sodium chloride by centrifugation.

C-26

Empirical formula from combustion data

Answer **C**

Question 26

A 0.300 g compound contains only carbon, hydrogen and oxygen. Complete combustion produces 0.440 g CO_2 and 0.180 g H_2O. What is the empirical formula? Take A_r: C=12, H=1, O=16.

A CHO

B C_2H_4O

C CH_2O

D CH_4O

E C_2H_2O

WORKED SOLUTION

0.440 g CO_2 =0.010 mol CO_2, so the sample contained 0.010 mol C =0.120 g C. 0.180 g H_2O =0.010 mol H_2O, so it contained 0.020 mol H =0.020 g H. Oxygen mass is 0.300−0.140=0.160 g =0.010 mol O. Ratio C:H:O =1:2:1, so CH_2O.

Therefore the correct option is **C**.

C-27

Gas volume and purity

Answer **A**

Question 27

A 10.0 g sample of impure calcium carbonate reacts with excess hydrochloric acid. A calibration shows that the gas-collection apparatus captures 75.0% of the carbon dioxide produced; the syringe reading is 1.26 dm^3 at rtp. Assume that calcium carbonate is the only substance in the sample that produces carbon dioxide.

$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$

Take $M_r(CaCO_3)=100$ and molar gas volume=24 $dm^3\ mol^{-1}$. What is the percentage purity?

- **A** 70.0%
- **B** 52.5%
- **C** 35.0%
- **D** 93.3%
- **E** 28.0%

WORKED SOLUTION

The actual carbon dioxide volume is 1.26/0.750=1.68 dm^3. This is 1.68/24=0.0700 mol CO_2, so the sample contained 0.0700 mol $CaCO_3$, with mass 0.0700×100=7.00 g. The percentage purity is (7.00/10.0)×100=70.0%.

Therefore the correct option is **A**.

Distractor check: B treats the captured volume as the full yield; C incorrectly applies the coefficient 2 from HCl; D corrects for the 75% capture twice; E uses $A_r(Ca)=40$ instead of $M_r(CaCO_3)=100$.

ANSWER KEY

ESAT 2024-25 Recall Mock - Biology

27 hard, recall-informed Biology questions with complete worked solutions.

Construction verification: the summary key, marked option, card answer and worked-solution answer have been matched for every question. This is original, unofficial ESAT-style practice informed by the published specification and broad public candidate impressions; it is not an official UAT-UK paper or a reproduction of live-test material.

Biology answer key

Q1 A	Q2 F	Q3 B	Q4 C	Q5 A	Q6 E	Q7 B	Q8 C	Q9 C
Q10 F	Q11 C	Q12 D	Q13 D	Q14 C	Q15 D	Q16 B	Q17 E	Q18 D
Q19 A	Q20 B	Q21 H	Q22 A	Q23 G	Q24 E	Q25 B	Q26 E	Q27 E

Biology

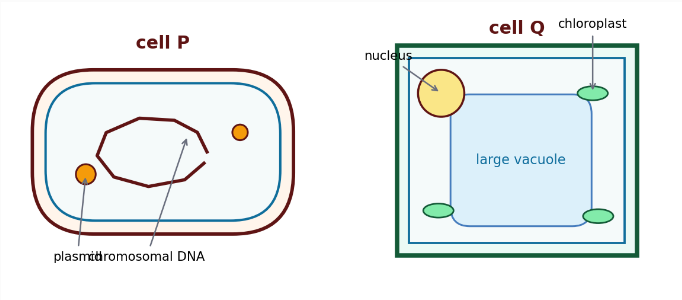
27 hard, recall-informed Biology questions with complete worked solutions.

B-01 | Cells and membrane transport | Approachable | Answer **A**

Question 1

The diagrams show two cells, P and Q.

Which conclusion is supported by the diagrams and established cell biology?



Programmatically generated diagram; not necessarily to scale.

- **A** P can respire without mitochondria and may contain plasmid DNA.
- **B** P cannot release energy because it has no mitochondria.
- **C** The cell wall of Q proves that Q cannot be a bacterial cell.
- **D** P has its chromosomal DNA enclosed by a nuclear membrane.
- **E** The presence of cytoplasm distinguishes Q from P.

WORKED SOLUTION

P is a bacterium. Bacteria carry out respiration using enzymes associated with their cell membrane and cytoplasm, despite lacking mitochondria, and they may contain both chromosomal and plasmid DNA. A cell wall does not uniquely identify a plant cell because bacterial cells also possess walls; both cells also contain cytoplasm.

The correct option is **A**.

Distractor check: B treats mitochondria as essential for all respiration; C treats a wall as uniquely plant-like; D invents a bacterial nucleus; E overlooks that both cells contain cytoplasm.

B-02

Osmosis and graph interpretation

Hard

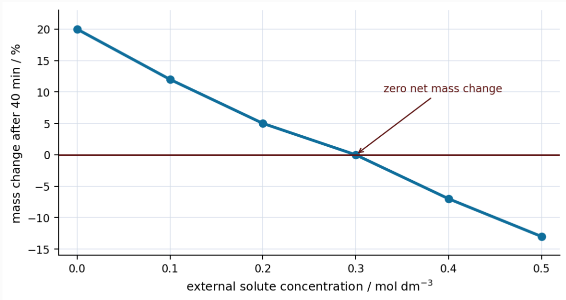
Answer **F**

Question 2

Identical plant-tissue cylinders are left for 40 minutes in solutions of different solute concentration. The graph shows their mean percentage change in mass. Assume that the cylinders have reached osmotic equilibrium by 40 minutes and that the solute does not cross cell membranes.

1. At 0.30 mol dm^{-3}, zero net mass change indicates that the tissue and the solution have the same water potential.
2. The graph shows that the instantaneous rate of osmosis at 40 minutes is greatest in 0.50 mol dm^{-3} solution.
3. A cylinder with initial mass 4.00 g would have final mass 3.48 g after a 13% decrease.

Which statements are correct?



Programmatically generated diagram; not necessarily to scale.

A none of them	**B** 1 only
C 2 only	**D** 3 only
E 1 and 2 only	**F** 1 and 3 only
G 2 and 3 only	**H** 1, 2 and 3

WORKED SOLUTION

Under the stated equilibrium and impermeant-solute assumptions, zero net mass change identifies the isotonic concentration, so statement 1 is correct. The graph gives total mass change after 40 minutes, not the instantaneous rate at that time, so statement 2 does not follow. For statement 3, $4.00\times0.87=3.48$ g.

Statements 1 and 3 only are correct, so the answer is **F**.

Distractor check: A large final mass change is not a measurement of the instantaneous rate at the finishing time.

B-03

Meiosis and chromosome number | Hard | Answer **B**

Question 3

A diploid cell has $2n = 8$. In normal meiosis I, homologous chromosomes separate; sister chromatids separate during meiosis II. Before meiosis, the cell's DNA replicates normally. During meiosis I, one homologous pair fails to separate: both homologues move to the same daughter cell. Meiosis II then proceeds normally.

Which outcome is expected?

- **A** four gametes with 4 chromosomes each; 16 chromosomes in total
- **B** two gametes with 5 chromosomes and two with 3; 16 chromosomes in total
- **C** two gametes with 5 chromosomes and two with 3; 8 chromosomes in total
- **D** one gamete with 5 chromosomes, one with 3 and two with 4; 16 chromosomes in total
- **E** four gametes with 8 chromosomes each; 32 chromosomes in total

WORKED SOLUTION

The normal haploid number is 4. Failure of one homologous pair to separate at meiosis I gives one cell an extra chromosome and the other one fewer. Normal meiosis II therefore produces two gametes with 5 chromosomes and two with 3. Their total is $2(5)+2(3)=16$.

The correct option is **B**.

Distractor check: A describes normal meiosis; D is the pattern expected from a different separation error affecting only one meiosis-II division.

B-04

Pedigrees and conditional probability

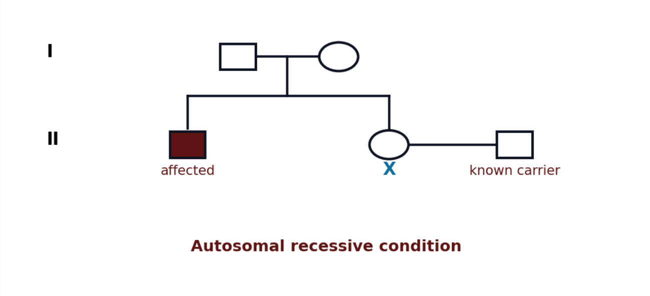
Very hard

Answer **C**

Question 4

The pedigree concerns an autosomal recessive condition. The two unaffected people in generation I have one affected child and one unaffected child, X. X has a partner who is known to be heterozygous.

What is the probability that *exactly one* of the first two children of X and the partner is affected?



Programmatically generated diagram; not necessarily to scale.

A 1/8

B 3/16

C 1/4

D 3/8

E 1/2

WORKED SOLUTION

The generation-I parents must both be carriers. Given that X is unaffected, X has conditional probability $2/3$ of being a carrier. If X is a carrier, the cross is $\mathrm{Aa} \times \mathrm{Aa}$, so the probability of exactly one affected child in two is

$$2 \times (1/4) \times (3/4) = 3/8.$$

Multiplying by the probability that X is a carrier gives $(2/3)(3/8){=}1/4$. The correct option is **C**.

Distractor check: D assumes X is certainly a carrier; B uses an unjustified one-half carrier probability.

B-05

DNA base pairing and triplets

Standard

Answer **A**

Question 5

One strand of a double-stranded DNA molecule contains 60 nucleotides:

base	A	T	G	C
number on this strand	12	9	15	24

Treat the strand shown as the coding strand, read from its first base in one continuous reading frame. Assume that the 60 bases form complete triplets and that none is a stop triplet. Which option gives:

- the percentage of all nucleotides in the double-stranded molecule that are adenine; and
- the maximum number of amino acids for which this section can code?

A 17.5%; 20

B 35%; 20

C 17.5%; 60

D 20%; 20

E 35%; 40

WORKED SOLUTION

The complementary strand contains 9 adenines, paired with the 9 thymines shown. Total adenine is therefore $12+9=21$ out of 120 nucleotides, or 17.5%. Under the stated coding-strand, reading-frame and no-stop assumptions, 60 bases form $60/3=20$ codons, so the section codes for 20 amino acids.

The correct option is **A**.

Distractor check: 20% uses adenine on the shown strand only; 35% incorrectly combines A and T and calls both adenine.

B-06

Mutation and the genetic code | Standard | Answer **E**

Question 6

A small part of a genetic code is supplied.

triplet	amino acid	triplet	amino acid
AAA	lysine	AAG	lysine
AAC	asparagine	GAA	glutamate
GAG	glutamate	GAC	aspartate

The original sequence is **AAA - GAA - AAC**. Three mutants have sequences:

P: AAG - GAA - AAC
Q: AAA - GAG - AAC
R: AAA - GAA - GAC

1. P specifies the same amino-acid sequence as the original.
2. Q specifies the same amino-acid sequence as the original.
3. R produces the same phenotype as the original.

Which statements are supported by the information supplied?

- **A** none of them
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

AAG and AAA both specify lysine, while GAG and GAA both specify glutamate. P and Q therefore specify the same three amino acids as the original. R changes asparagine to aspartate, and the code table gives no phenotype evidence, so statement 3 is not supported.

Statements 1 and 2 only are correct, so the answer is **E**.

Distractor check: A base substitution can be silent, while an amino-acid change cannot by itself establish the resulting phenotype.

B-07

Stem cells and selective breeding

Standard

Answer **B**

Question 7

Which statement correctly compares stem-cell potential with a possible consequence of repeated selective breeding?

- **A** Pluripotent cells can form a complete organism, and selective breeding necessarily increases genetic variation.
- **B** Totipotent cells can potentially form a complete organism, and repeated selection can reduce genetic variation.
- **C** Totipotent and pluripotent cells have identical potential, and selection leaves allele frequencies unchanged.
- **D** Only pluripotent cells can form specialised cells, and selective breeding always creates new alleles.
- **E** Totipotent cells form fewer cell types than pluripotent cells, and selection prevents inheritance.

WORKED SOLUTION

Totipotent cells can generate all cell types required for a complete organism. Pluripotent cells can form many body cell types but do not have the same complete developmental potential. Repeated selective breeding changes allele frequencies and may remove alleles not associated with the selected phenotype, reducing variation.

The correct option is **B**.

Distractor check: The alternatives reverse totipotent and pluripotent potential or claim that selection necessarily creates variation rather than sometimes reducing it.

B-08

Natural selection and antibiotic resistance

Standard

Answer **C**

Question 8

Before antibiotic treatment, a bacterial population contains 20 000 cells. Of these, 5% carry an inherited resistance variant. The treatment kills 99% of susceptible cells and 10% of resistant cells.

What fraction of the surviving population carries the resistance variant?

- **A** 1/20
- **B** 9/200
- **C** 90/109
- **D** 99/100
- **E** 18/19

WORKED SOLUTION

There are initially 1000 resistant and 19 000 susceptible cells. Survivors are $0.90(1000)=900$ resistant and $0.01(19000)=190$ susceptible. The required fraction is

$$900/(900+190) = 90/109.$$

The correct option is **C**.

Distractor check: 1/20 is the initial proportion; 18/19 ignores the different survival rate of susceptible cells.

B-09

Enzymes and pH data

Standard

Answer **C**

Question 9

Two enzymes are tested at the same temperature with substrate in excess.

pH	4	5	6	7	8
initial rate of A	4	12	16	16	6
initial rate of B	2	6	12	20	10

1. The equal measured rates at pH 6 and pH 7 identify pH 6.5 as the optimum pH of A.
2. At pH 7, the rate for B is 25% greater than the rate for A.
3. The higher rate of B at pH 8 than at pH 4 shows that B retains its normal three-dimensional shape at pH 8.

Which statements are correct?

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Equal rates at the two tested pH values do not locate the optimum between them, so statement 1 is not supported. At pH 7, the increase from 16 to 20 is $4/16=25\%$, so statement 2 is correct. A non-zero rate does not show that every enzyme molecule retains its normal shape, so statement 3 does not follow.

Statement 2 only is correct, so the answer is **C**.

Distractor check: Discrete measurements cannot locate an untested optimum, and residual activity is not evidence about every enzyme molecule.

B-10

Enzyme progress curves

Hard

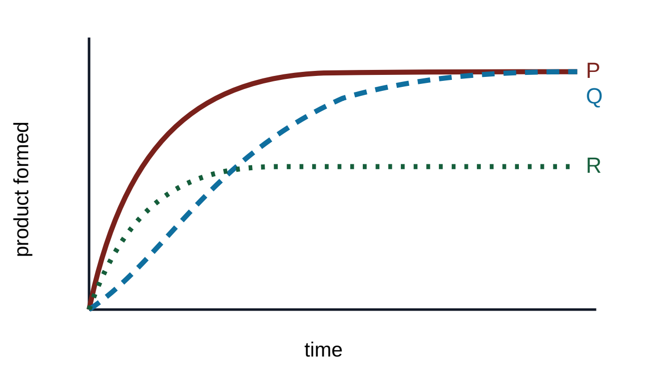
Answer **F**

Question 10

Three experiments use identical initial amounts of enzyme and substrate. Only temperature differs. The graph shows product formed.

1. P has a greater initial rate than Q.
2. The common final product amount for P and Q shows that enzyme molecules remained structurally unchanged in both experiments.
3. R is consistent with enzyme becoming inactive before all the substrate is converted.

Which statements are correct?



A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Initial rate is the initial gradient, which is steeper for P than for Q. Equal final amounts do not show that every enzyme molecule remained unchanged; both experiments can still eventually convert the same finite amount of substrate. R rises rapidly but stops at a lower product amount despite the identical start, consistent with loss of functioning active sites before completion.

Statements 1 and 3 only are correct, so the answer is **F**.

Distractor check: A plateau records product amount, not the molecular history of the enzyme population.

B-11

Aerobic and anaerobic respiration

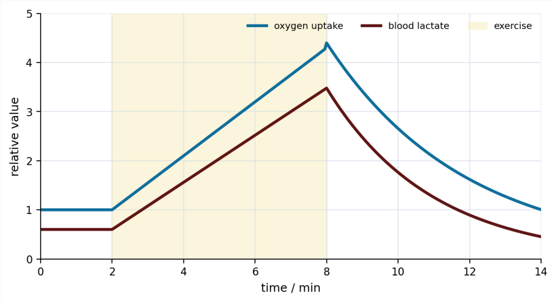
Standard

Answer **C**

Question 11

The graph shows oxygen uptake and blood lactate before, during and after exercise.

Which conclusion is best supported by the graph?



Programmatically generated diagram; not necessarily to scale.

A Only anaerobic respiration occurred during exercise because lactate increased.

B Only aerobic respiration occurred because oxygen uptake increased.

C Both aerobic and anaerobic respiration occurred, but the graph does not show that all post-exercise oxygen was used solely to remove lactate.

D The return of lactate to near its initial value proves that oxygen had no other post-exercise role.

E The graph shows that lactate was converted directly into oxygen after exercise.

WORKED SOLUTION

Rising lactate supports the occurrence of some anaerobic respiration, while oxygen uptake during exercise is consistent with aerobic respiration occurring as well. The graph does not partition post-exercise oxygen use among recovery processes, so it cannot show that all of that oxygen was used solely for lactate removal.

The correct option is **C**.

Distractor check: A and B treat the two pathways as mutually exclusive; D overclaims from correlation; E invents a direct conversion not represented in the graph.

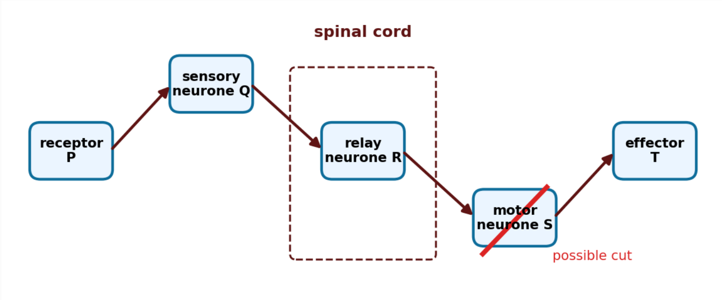
B-12

Reflex arcs | Approachable | Answer **D**

Question 12

The diagram represents a withdrawal reflex. Motor neurone S is severed between the spinal cord and effector T; all other structures remain functional.

Which outcome is most likely immediately after the same stimulus is applied?



Programmatically generated diagram; not necessarily to scale.

A P cannot detect the stimulus, but T contracts

B impulses do not reach the spinal cord, but T contracts

C impulses reach T, but not the spinal cord

D sensory impulses reach the spinal cord, but T does not contract

E relay neurone R becomes the effector

WORKED SOLUTION

The receptor, sensory neurone and relay neurone remain intact, so the sensory signal can reach the central nervous system. The severed motor pathway cannot carry the output to the effector, so T does not contract.

The correct option is **D**.

Distractor check: The sensory and motor limbs of the reflex must be considered separately.

B-13

Ventilation and graph interpretation

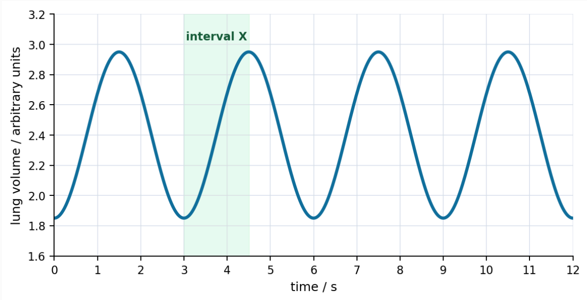
Standard

Answer **D**

Question 13

The graph shows lung volume over 12 seconds. Assume the pattern continues.

Which option correctly gives the breathing rate and describes interval X?



Programmatically generated diagram; not necessarily to scale.

A 5 breaths min^{-1}; diaphragm relaxes and thoracic pressure falls

B 20 breaths min^{-1}; diaphragm relaxes and thoracic pressure rises

C 15 breaths min^{-1}; diaphragm contracts and thoracic pressure rises

D 20 breaths min^{-1}; diaphragm contracts and thoracic pressure falls

E 4 breaths min^{-1}; diaphragm relaxes and thoracic pressure is unchanged

WORKED SOLUTION

One complete cycle lasts 3 s, so the rate is $60/3=20$ breaths per minute. During X, lung volume rises: this is inspiration, when the diaphragm contracts and thoracic pressure falls below atmospheric pressure.

The correct option is **D**.

Distractor check: Counting peaks rather than complete peak-to-peak cycles can produce an incorrect rate.

B-14

Cardiac output and stroke volume

Standard

Answer **C**

Question 14

A person's heart data are shown.

condition	heart rate / beats min^{-1}	cardiac output / dm^3 min^{-1}
rest	75	5.25
exercise	150	15.0

The volume pumped per beat is called the *stroke volume*. Which option gives the stroke volume during exercise and the percentage increase in stroke volume from rest to exercise?

A 100 cm^3; 50%

B 70 cm^3; about 43%

C 100 cm^3; about 43%

D 150 cm^3; 100%

E 35 cm^3; 30%

WORKED SOLUTION

At rest, stroke volume is $5250/75=70$ cm^3. During exercise it is $15000/150=100$ cm^3. The increase is

$$(100\text{-}70)/70 \times 100\% = 300/7\% \approx 43\%.$$

The correct option is **C**.

Distractor check: A divides the 30 cm^3 increase by the wrong reference value; B reports the resting stroke volume.

B-15

ADH and kidney response

Standard

Answer **D**

Question 15

A dehydrated patient releases ADH normally, but receptor molecules in the collecting ducts do not respond to ADH.

Compared with a healthy dehydrated person, which combination is most likely?

- **A** low blood ADH; high water reabsorption; small volume of concentrated urine
- **B** high blood ADH; high water reabsorption; small volume of concentrated urine
- **C** low blood ADH; low water reabsorption; large volume of dilute urine
- **D** high blood ADH; low water reabsorption; large volume of dilute urine
- **E** high blood ADH; low water reabsorption; small volume of concentrated urine

WORKED SOLUTION

Dehydration lowers blood water potential, so the normal endocrine response is high ADH. If the collecting ducts cannot respond, ADH cannot produce its normal increase in water reabsorption. More water therefore remains in a larger volume of relatively dilute urine.

The correct option is **D**.

Distractor check: B describes a healthy response; C confuses failure of the target organ with failure to secrete the hormone.

B-16

Blood-glucose homeostasis

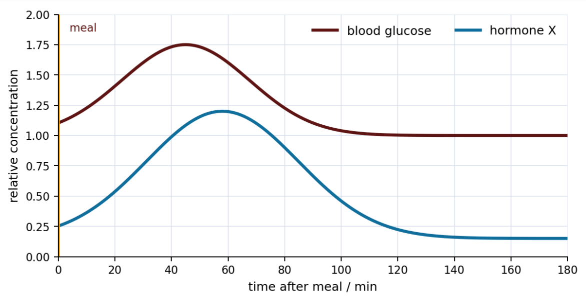
Approachable

Answer **B**

Question 16

The graph shows blood glucose and the concentration of hormone X after a carbohydrate-rich meal. A different hormone, Y, rises during prolonged fasting and helps raise blood glucose.

Which identification and action are correct?



Programmatically generated diagram; not necessarily to scale.

A X is glucagon and promotes a fall in blood glucose

B X is insulin and promotes a fall in blood glucose

C X is adrenaline and Y is insulin

D X is insulin and Y also promotes a fall in blood glucose

E X is glucagon and Y is ADH

WORKED SOLUTION

Hormone X rises after blood glucose rises and acts in the negative-feedback response that brings glucose down, so X is insulin. The hormone that helps raise blood glucose during fasting is glucagon.

The correct option is **B**.

Distractor check: The graph shows association in time, while the stem supplies the homeostatic role needed to distinguish insulin from glucagon.

B-17

Reproductive hormones

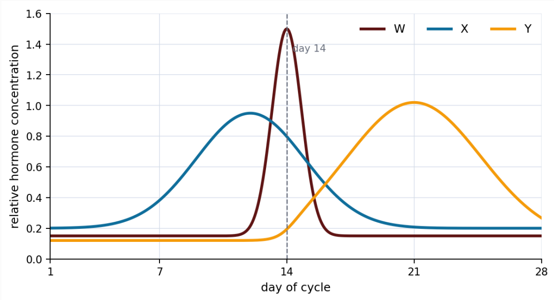
Standard

Answer **E**

Question 17

The graph shows three hormone patterns during a menstrual cycle.

Which identification of the three hormone patterns is most consistent with the graph?



Programmatically generated diagram; not necessarily to scale.

A W: FSH; X: progesterone; Y: oestrogen

B W: progesterone; X: oestrogen; Y: LH

C W: LH; X: progesterone; Y: oestrogen

D W: oestrogen; X: LH; Y: progesterone

E W: LH; X: oestrogen; Y: progesterone

WORKED SOLUTION

The sharp mid-cycle surge is characteristic of LH, so W is LH. Oestrogen rises before ovulation, matching X. Progesterone is elevated mainly after ovulation and helps maintain the uterine lining, matching Y.

The correct option is **E**.

Distractor check: The peak shapes and timing, rather than the absolute vertical scale, identify the hormones.

B-18

Vaccination data

Standard

Answer **D**

Question 18

In a randomised trial, 18 of 600 vaccinated participants and 60 of 500 control participants developed a disease during the same follow-up period.

What is the proportional reduction in disease risk in the vaccinated group compared with the control group?

- **A** 3%
- **B** 9%
- **C** 70%
- **D** 75%
- **E** 85%

WORKED SOLUTION

The risks are $18/600=3\%$ and $60/500=12\%$. The proportional reduction is

$$(12\%-3\%)/12\% = 3/4 = 75\%.$$

The correct option is **D**.

Distractor check: The unequal group sizes make comparing the raw case counts invalid; 9% is the absolute percentage-point reduction.

B-19

Clinical testing and bias

Approachable

Answer **A**

Question 19

Which statement gives the most justified interpretation of clinical-trial design?

- **A** Random allocation reduces systematic group differences, double blinding reduces expectation and observer bias, and short follow-up cannot establish lifelong safety.
- **B** Random allocation guarantees identical groups and therefore removes all confounding variables.
- **C** Double blinding changes the biological action of the medicine and prevents adverse effects.
- **D** No immediate adverse effect proves that the medicine is safe for lifelong use.
- **E** Random allocation and double blinding serve the same purpose, so using both provides no additional control.

WORKED SOLUTION

Random allocation reduces systematic differences between groups but does not guarantee perfectly identical groups. Double blinding addresses expectation and observer bias. A short trial cannot establish effects that might appear only after prolonged use.

The correct option is **A**.

Distractor check: B and E overstate or merge distinct controls; C invents a physiological effect of blinding; D ignores the limit imposed by trial duration.

B-20

Belt transects and population density

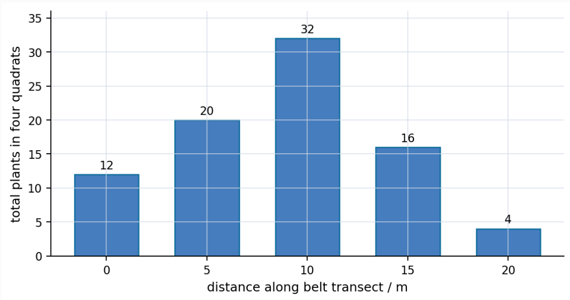
Standard

Answer **B**

Question 20

At each distance on a belt transect, four quadrats of area 0.25 m^2 are sampled. The graph shows the total count in the four quadrats.

Assuming an area of 12.5 m^2 has the same mean density as the samples at 10 m, which option gives the density at 10 m and the estimated number of plants in that area?



Programmatically generated diagram; not necessarily to scale.

- **A** 8 m^{-2}; 100
- **B** 32 m^{-2}; 400
- **C** 32 m^{-2}; 800
- **D** 128 m^{-2}; 1600
- **E** 8 m^{-2}; 400

WORKED SOLUTION

The four quadrats together sample $4(0.25)=1.00\ \mathrm{m}^2$. A count of 32 therefore gives a density of $32\ \mathrm{m}^{-2}$. The estimated population is $32(12.5)=400$.

The correct option is **B**.

Distractor check: A divides the count by four but forgets that each quadrat is only one-quarter of a square metre; D multiplies instead of combining areas.

B-21

Energy flow and decomposers

Hard

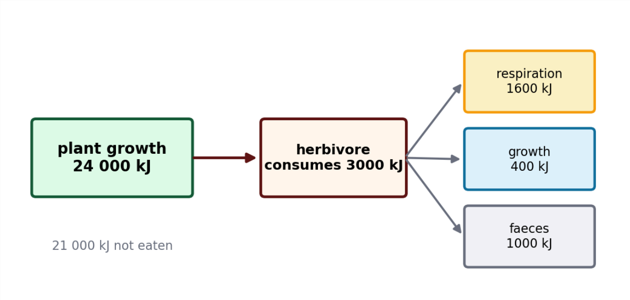
Answer **H**

Question 21

The diagram shows annual energy flow from 24 000 kJ of plant growth. A herbivore consumes 3000 kJ; 1000 kJ leaves in faeces, 1600 kJ is transferred by respiration and 400 kJ becomes new herbivore biomass. For this question, **absorbed energy** means consumed energy minus energy lost in faeces.

1. The energy absorbed by the herbivore is 2000 kJ.
2. Twenty per cent of the absorbed energy becomes new herbivore biomass.
3. If all uneaten plant material and faeces enter decomposer food chains, they receive 11/12 of the plant-growth energy shown.

Which statements are correct?



Programmatically generated diagram; not necessarily to scale.

- **A** none of them
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

Using the definition in the stem, absorbed energy is $3000-1000=2000$ kJ. Growth efficiency relative to absorbed energy is $400/2000=20\%$. Uneaten plant material is $24000-3000=21000$ kJ; adding faeces gives 22 000 kJ, and $22000/24000=11/12$.

All three statements are correct, so the answer is **H**.

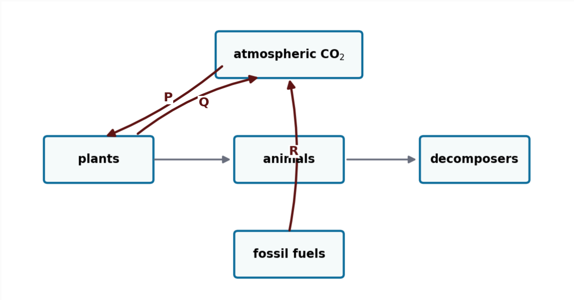
Distractor check: The denominator changes between consumption and absorbed-energy efficiencies; the energy-flow diagram must be read before calculating.

B-22

Carbon cycle | Standard | Answer **A**

Question 22

The diagram shows part of the carbon cycle. Which row correctly identifies processes P, Q and R?



Programmatically generated diagram; not necessarily to scale.

A P photosynthesis; Q respiration; R combustion

B P respiration; Q photosynthesis; R decomposition

C P transpiration; Q respiration; R combustion

D P photosynthesis; Q translocation; R respiration

E P combustion; Q photosynthesis; R decomposition

WORKED SOLUTION

P transfers carbon dioxide from the atmosphere into plant biomass by photosynthesis. Q returns carbon from organisms to atmospheric carbon dioxide by respiration. R transfers carbon from fossil fuels to atmospheric carbon dioxide by combustion.

The correct option is **A**.

Distractor check: The arrow direction is decisive: respiration and photosynthesis move carbon in opposite directions.

B-23

Eutrophication and experimental evidence

Standard

Answer **G**

Question 23

Three identical lake microcosms are maintained for two weeks. Assume that the decomposer inhibitor does not directly affect algal growth, light transmission or oxygen solubility.

treatment	algal biomass / units	light at submerged plants / %	dissolved oxygen / mg dm^{-3}
control	2	80	8
nitrate added	10	25	3
nitrate + decomposer inhibitor	9	28	7

1. The low light in the nitrate treatment is mainly caused by decomposer activity.
2. Under these conditions, added nitrate is associated with increased algal biomass.
3. Restoring much of the oxygen without restoring much light supports the conclusion that decomposition contributes to oxygen loss.

Which statements are correct?

- **A** none of them
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

Blocking decomposers barely changes algal biomass or light, so statement 1 is false: decomposers are not the main explanation for shading. Nitrate addition is followed by a fivefold algal increase, so statement 2 is supported. The inhibitor raises oxygen from 3 to 7 mg dm^{-3} while light remains low, supporting statement 3 and a role for decomposer respiration in oxygen loss.

Statements 2 and 3 only are correct, so the answer is **G**.

Distractor check: The controlled comparison is nitrate versus nitrate plus inhibitor, not nitrate versus control alone.

B-24

Photosynthesis and limiting factors

Hard

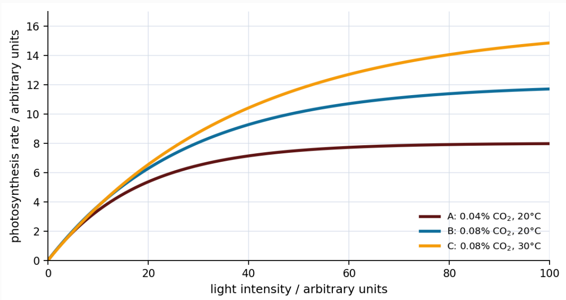
Answer **E**

Question 24

The graph shows photosynthesis rate under three combinations of carbon-dioxide concentration and temperature.

1. At low light intensity, light is a limiting factor in all three conditions.
2. At high light intensity and 20°C, the comparison of A with B shows carbon dioxide limiting the rate in A.
3. Raising the temperature from 20°C to 30°C produces the same absolute increase in photosynthesis rate at every light intensity shown.

Which statements are correct?



Programmatically generated diagram; not necessarily to scale.

A none of them	**B** 1 only
C 2 only	**D** 3 only
E 1 and 2 only	**F** 1 and 3 only
G 2 and 3 only	**H** 1, 2 and 3

WORKED SOLUTION

All curves initially rise with light, showing light limitation. At high light, B exceeds A when only carbon dioxide concentration has changed, so carbon dioxide limits A there. Statement 3 is false: the vertical separation between B and C varies with light intensity and is zero at zero light, so the temperature increase does not produce the same absolute rate increase throughout.

Statements 1 and 2 only are correct, so the answer is **E**.

Distractor check: The size of a temperature effect must be read from the vertical separation of B and C; it is not constant across the graph.

B-25

Xylem, phloem and girdling

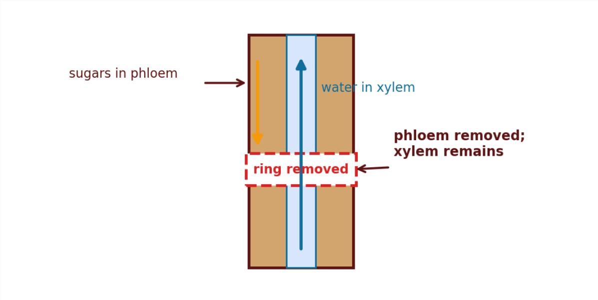
Standard

Answer **B**

Question 25

A narrow ring of tissue containing phloem is removed from a woody stem. The xylem remains intact.

Which outcome is most likely?



Programmatically generated diagram; not necessarily to scale.

A Upward water movement stops immediately, and sugars accumulate below the ring.

B Upward water movement may continue initially, sugars accumulate above the ring, and root respiration may later decrease.

C Phloem continues carrying sugars to the roots because the xylem is intact.

D Sugars accumulate below the ring, causing an immediate increase in root growth.

E Neither transport system is affected because both xylem and phloem lie deeper than the removed ring.

WORKED SOLUTION

The intact xylem can continue transporting water and mineral ions upward immediately after girdling. Removing phloem interrupts translocation from the leaves toward the roots, so sugars accumulate above the ring and the roots later receive less respiratory substrate.

The correct option is **B**.

Distractor check: Xylem and phloem directions/functions must not be interchanged, and the word 'immediately' matters.

B-26

Restriction enzymes and ligase

Standard

Answer **E**

Question 26

A circular plasmid has one recognition site for restriction enzyme E. A linear donor DNA molecule carries gene G and has an E recognition site at each end. Both DNA molecules are completely cut with E and then mixed with DNA ligase. Assume that every E cut produces compatible sticky ends.

1. Cutting the plasmid once converts it into one linear DNA molecule.
2. A donor fragment carrying G can become joined into the plasmid.
3. The donor fragment can join into the plasmid in only one orientation.

Which statements are correct?

- **A** none of them
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

A single cut opens a circular plasmid to give one linear DNA molecule, so statement 1 is correct. Compatible sticky ends allow the donor fragment to ligate into the opened plasmid, so statement 2 is correct. Because the same compatible end is present at each end of the donor, it can be inserted in either orientation; statement 3 is false.

Statements 1 and 2 only are correct, so the answer is **E**.

Distractor check: Compatible sticky ends permit joining but do not by themselves impose a direction on the insert.

B-27

Photosynthesis experiment and inverse proportion

Very hard

Answer **E**

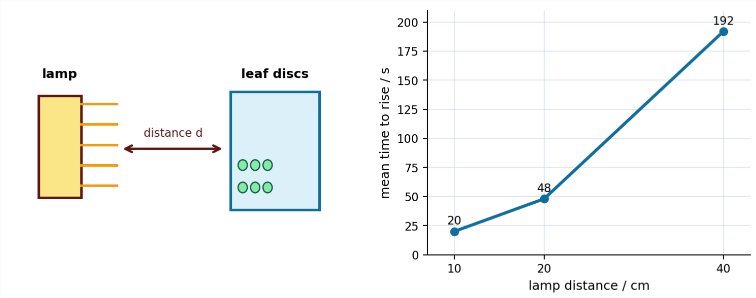
Question 27

Identical leaf discs are placed in identical solutions at constant temperature and carbon-dioxide concentration. Mean time to rise is used as an inverse measure of photosynthesis rate. Light intensity from the lamp is proportional to $1/d^2$, where d is lamp distance.

The observed mean times are 20 s at 10 cm, 48 s at 20 cm and 192 s at 40 cm.

1. If light alone limits the rate between 20 cm and 40 cm, the predicted mean time at 30 cm is 108 s.
2. The inverse-square prediction from 20 cm to 10 cm is 12 s, so the observed 20 s is consistent with another factor becoming limiting at high light intensity.
3. Raising the temperature would reduce the 10 cm mean time to the inverse-square prediction of 12 s.

Which statements are correct?



Programmatically generated diagram; not necessarily to scale.

- **A** none of them
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

Because rate is proportional to $1/\text{time}$ and light to $1/d^2$, time is proportional to d^2 while light is the only limiting factor. Thus $48(30/20)^2=108$ s. Moving from 20 cm to 10 cm predicts $48(10/20)^2=12$ s, but the observed time is longer; another factor could be limiting. The data do not identify temperature as that factor or determine the result of changing it.

Statements 1 and 2 only are correct, so the answer is **E**.

Distractor check: The calculation tests inverse proportionality; the interpretation distinguishes a possible explanation from one established by the data.